Milestone 2: Dataset Preparation

Socratic AI Tutor

Group 2 | Data Science and AI Lab, IIT Madras BS

Sandeep Nalla, Hariom Kumar, Hariom Pandey, Mrityunjay Chakraborty

What we prepared

Dataset	Used for	Size after preparation
ASSISTments 2009	Learner model (DKT)	282,619 answers, 3,852 students
MathDial	Socratic model fine-tuning	18,541 tutor replies
NCERT Maths + Science	Course pack (study text)	3,164 text chunks, 10 books

All three are public. Raw files are not committed; scripts in data/ rebuild everything.

ASSISTments: learner model data

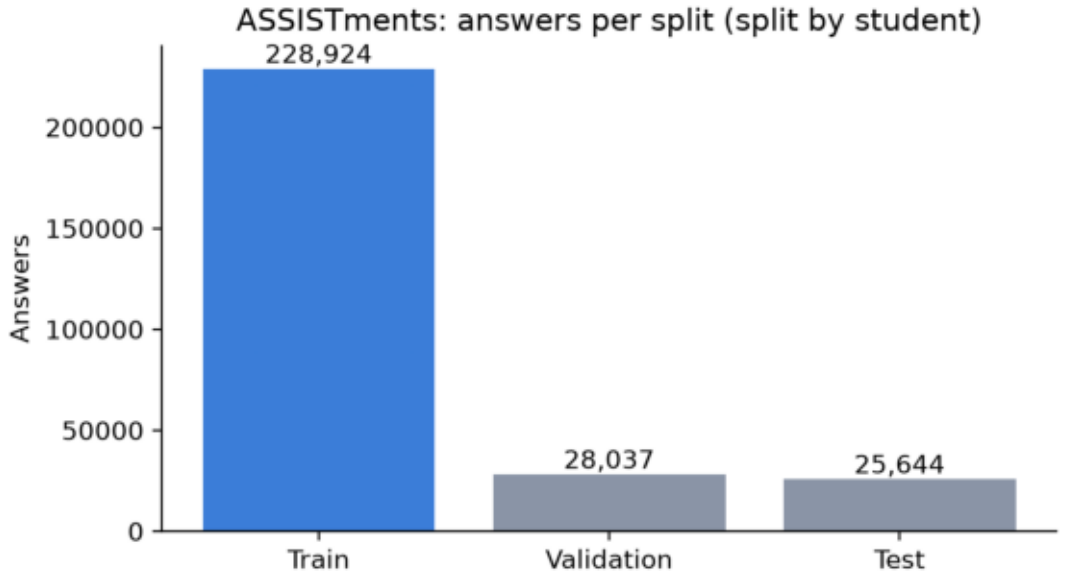
- Raw: 346,860 answers from 4,217 students

- Removed 63,755 answers with no skill label

- Dropped students with fewer than 3 answers (pyKT recipe)

- 149 skills, histories cut into windows of up to 200 answers

- Split by student: 80% train, 10% validation, 10% test



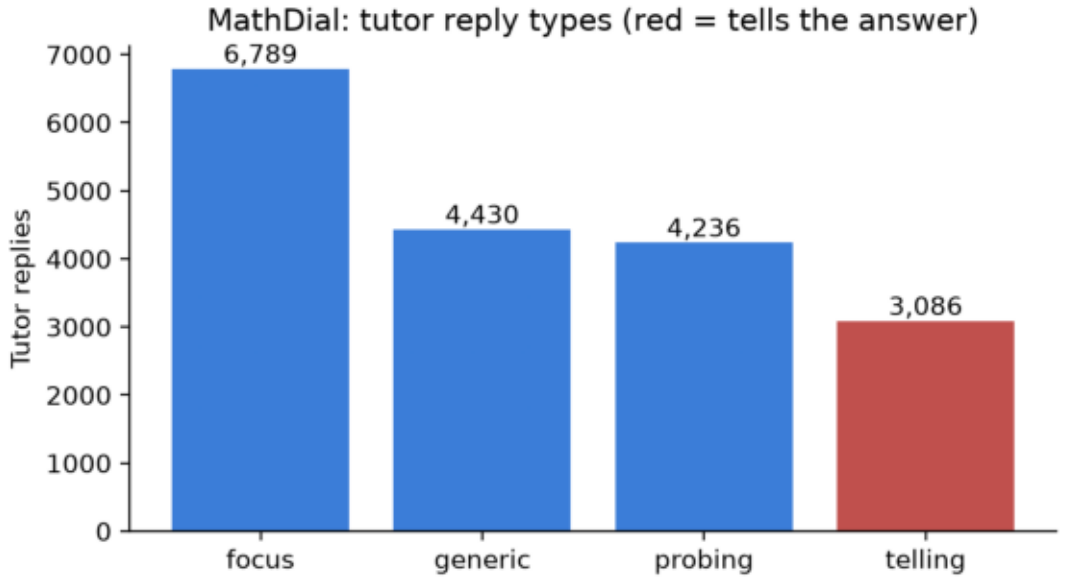
MathDial: Socratic model data

- One example per tutor reply: problem + conversation so far -> next reply

- Each reply keeps MathDial's label: probing, focus, generic, telling

- 'Telling' replies and replies containing the final answer are flagged and not used as training targets

- Validation split by problem, so a problem never sits in both train and validation



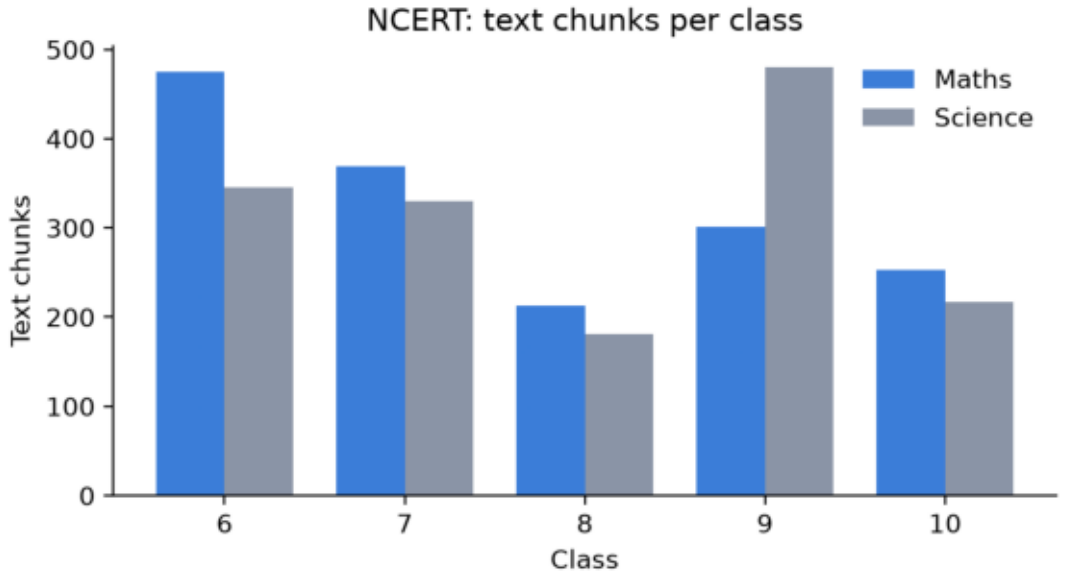
NCERT: study text

- Maths and Science, classes 6 to 10 (classes act as levels)

- 2,401 pages cleaned and cut into chunks of up to 250 words

- Each chunk keeps subject, class, chapter and page, so the tutor can point to the exact page

- Text stays local: the books are public but not ours to redistribute



Course packs: concepts and questions

- Hand-written chains of concepts that build across levels

- Example: percentages -> fractions -> multiplication and division

- Each concept links to its NCERT chapter, so the tutor can point to the page

- Practice questions have a hint and difficulty; answers are computed in code

Pack	Concepts	Prerequisite links	Questions
NCERT Maths	10	13	52
NCERT Science	6	6	18

Our own test scenarios

- Backtracking: 82 cases (correct, careless, hint, gap one or two levels down)

- Misconceptions: 32 wrong answers with the error behind each

- Guardrails: 45 messages (on-topic, off-topic, answer request)

- Never used for training or tuning; a second member reviews every label

Checks and limits

- 23 automatic checks pass: no student or problem shared across splits, valid graphs, consistent labels

- ASSISTments is US data and has no session information

- 283 of 394 MathDial test problems also appear in train (MathDial's own split)

- NCERT text: some digits lost in older books; study passages need a manual pass

Next steps

- Second member reviews concepts, prerequisites and scenarios

- Train DKT on ASSISTments and compare with the published baseline (AUC 0.754)

- Fine-tune the Socratic model on MathDial plus our NCERT examples

- Add the TA's test samples to our scenarios